

# Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

## Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in **R** using the `incumbents_subset.csv` dataset. Include all of your code.

## Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

**Answer:** A bivariate linear regression was estimated to assess the relationship between incumbent `voteshare` and the explanatory variable is `difflog`.

Table 1: Regression of Vote Share on Logged Spending Difference

| <i>Dependent variable:</i>               |                             |
|------------------------------------------|-----------------------------|
|                                          | voteshare                   |
| difflog                                  | 0.042***<br>(0.001)         |
| Constant                                 | 0.579***<br>(0.002)         |
| Observations                             | 3,193                       |
| R <sup>2</sup>                           | 0.367                       |
| Adjusted R <sup>2</sup>                  | 0.367                       |
| Residual Std. Error                      | 0.079 (df = 3191)           |
| F Statistic                              | 1,852.791*** (df = 1; 3191) |
| <i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 |                             |

Computed in R as:

```
1 #Q1.1: Run regression
2 modell <- lm(voteshare ~ difflog, data = inc_sub)
3 summary(modell)
4
5 #display output in LaTeX
6 stargazer(modell, type = "latex",
7           title = "Regression of Vote Share on Logged Spending Difference",
8           label = "tab:modell",
9           style = "default",
10          digits = 3,
11          header = FALSE)
```

2. Make a scatterplot of the two variables and add the regression line.

**Answer:** A scatterplot was generated to visualise the relationship between incumbent `voteshare` and `difflog`, with a fitted regression line overlaid to illustrate the model's linear trend.

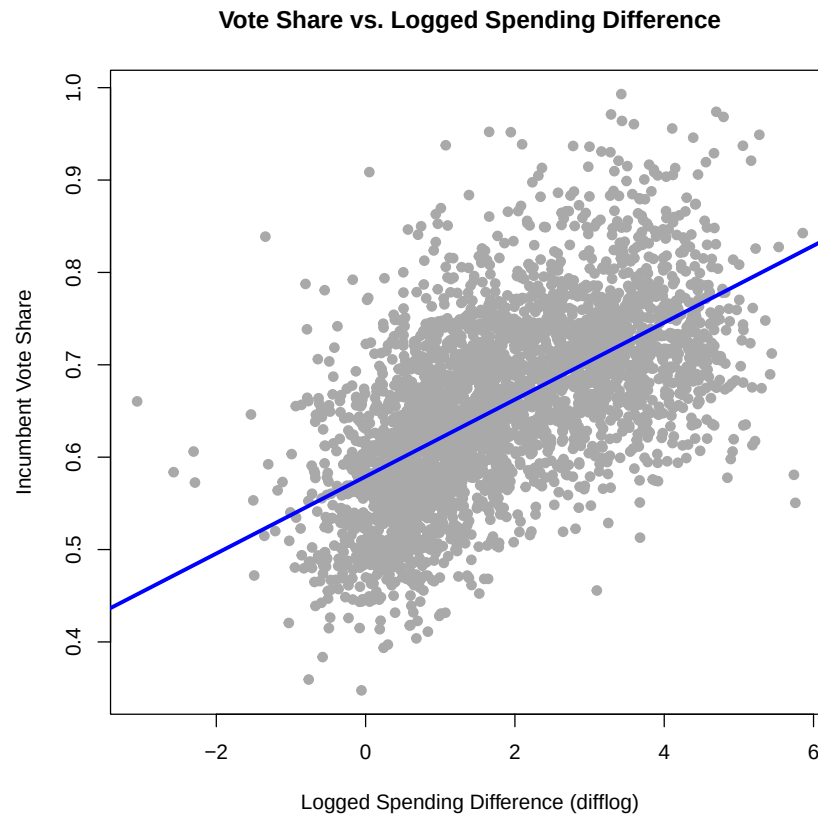


Figure 1: Scatterplot of incumbent vote share vs. logged spending difference

Computed in R as:

```
1 #Q1.2: Scatterplot with regression line
2 pdf("scatterplot_model1.pdf")
3 plot(inc_sub$difflog, inc_sub$voteshare,
4       main = "Vote Share vs. Logged Spending Difference",
5       xlab = "Logged Spending Difference (difflog)",
6       ylab = "Incumbent Vote Share",
7       pch = 19, col = "darkgray")
8
9 abline(model1, col = "blue", lwd = 3)
10 dev.off()
```

3. Save the residuals of the model in a separate object.

**Answer:** The residuals represent the portion of variation in incumbent `voteshare` that is not explained by `difflog`, capturing individual deviations from the model's predicted values. The residuals from the fitted regression model were saved into a separate object.

Computed in R as:

```
1 #Q1.3: Save residuals
2 residuals_model1 <- residuals(model1)
3 residuals_model1
```

4. Write the prediction equation.

**Answer:** The prediction equation indicates that for each one-unit increase in the logged spending difference favoring the incumbent, their expected vote share increases by approximately 4.2 percentage points. The prediction equation reads as follows.

$$\widehat{\text{voteshare}} = 0.579 + 0.042 \cdot \text{difflog}$$

## Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

**Answer:** A bivariate linear regression was estimated to assess the relationship between `presvote` and the explanatory variable is `difflog`.

Table 2: Regression of Presidential Vote Share on Logged Spending Difference

|                         | <i>Dependent variable:</i>  |
|-------------------------|-----------------------------|
|                         | presvote                    |
| difflog                 | 0.024***<br>(0.001)         |
| Constant                | 0.508***<br>(0.003)         |
| Observations            | 3,193                       |
| R <sup>2</sup>          | 0.088                       |
| Adjusted R <sup>2</sup> | 0.088                       |
| Residual Std. Error     | 0.110 (df = 3191)           |
| F Statistic             | 307.715*** (df = 1; 3191)   |
| Note:                   | *p<0.1; **p<0.05; ***p<0.01 |

Computed in R as:

```
1 #Q2.1: Run regression
2 model2 <- lm(presvote ~ difflog, data = inc_sub)
3 summary(model2)
4 #display output in LaTeX
5 stargazer(model2, type = "latex",
6           title = "Regression of Presidential Vote Share on Logged
7             Spending Difference",
8             label = "tab:model2",
9             style = "default",
10            digits = 3,
11            header = FALSE)
```

2. Make a scatterplot of the two variables and add the regression line.

**Answer:** A scatterplot was generated to visualise the relationship between `presvote` and `difflog`, with a fitted regression line overlaid to illustrate the model's linear trend.

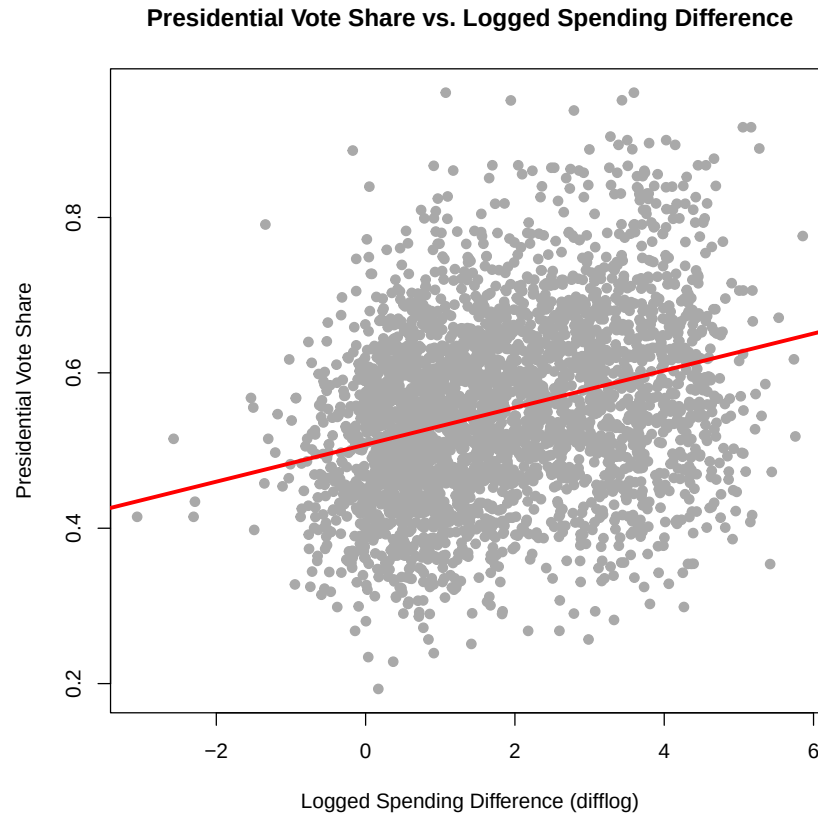


Figure 2: Scatterplot of incumbent vote share vs. logged spending difference

Computed in R as:

```
1 #Q2.2: Scatterplot with regression line
2 pdf("scatterplot_model2.pdf")
3 plot(inc_sub$difflog, inc_sub$presvote,
4       main = "Presidential Vote Share vs. Logged Spending Difference",
5       xlab = "Logged Spending Difference (difflog)",
6       ylab = "Presidential Vote Share",
7       pch = 19, col = "darkgray")
8
9 abline(model2, col = "red", lwd = 3)
10 dev.off()
```

3. Save the residuals of the model in a separate object.

**Answer:** The residuals represent the portion of variation in `presvote` that is not explained by `difflog`, capturing individual deviations from the model's predicted values. The residuals from the fitted regression model were saved into a separate object.

Computed in R as:

```
1 #Q1.3: Save residuals
2 residuals_model2 <- residuals(model2)
3 residuals_model2
```

4. Write the prediction equation.

**Answer:** The prediction equation suggests that for each one-unit increase in the logged spending difference favoring the incumbent, the expected presidential vote share for the incumbent's party increases by approximately 2.4 percentage points. The prediction equation reads as follows.

$$\widehat{\text{presvote}} = 0.508 + 0.024 \cdot \text{difflog}$$

### Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

**Answer:** A bivariate linear regression was estimated to assess the relationship between incumbent `voteshare` and the explanatory variable is `presvote`.

Table 3: Regression of Incumbent Vote Share on Presidential Vote Share

| <i>Dependent variable:</i>               |                           |
|------------------------------------------|---------------------------|
|                                          | voteshare                 |
| presvote                                 | 0.388***<br>(0.013)       |
| Constant                                 | 0.441***<br>(0.008)       |
| Observations                             | 3,193                     |
| R <sup>2</sup>                           | 0.206                     |
| Adjusted R <sup>2</sup>                  | 0.206                     |
| Residual Std. Error                      | 0.088 (df = 3191)         |
| F Statistic                              | 826.950*** (df = 1; 3191) |
| <i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 |                           |

Computed in R as:

```
1 #Q3.1: Run regression
2 model3 <- lm(voteshare ~ presvote, data = inc_sub)
3 summary(model3)
4 #display output in LaTeX
5 stargazer(model3, type = "latex",
6           title = "Regression of Incumbent Vote Share on Presidential
7             Vote Share",
8             label = "tab:model3",
9             style = "default",
10            digits = 3,
11            header = FALSE)
```

2. Make a scatterplot of the two variables and add the regression line.

**Answer:** A scatterplot was generated to visualise the relationship between incumbent `voteshare` and `presvote`, with a fitted regression line overlaid to illustrate the model's linear trend.

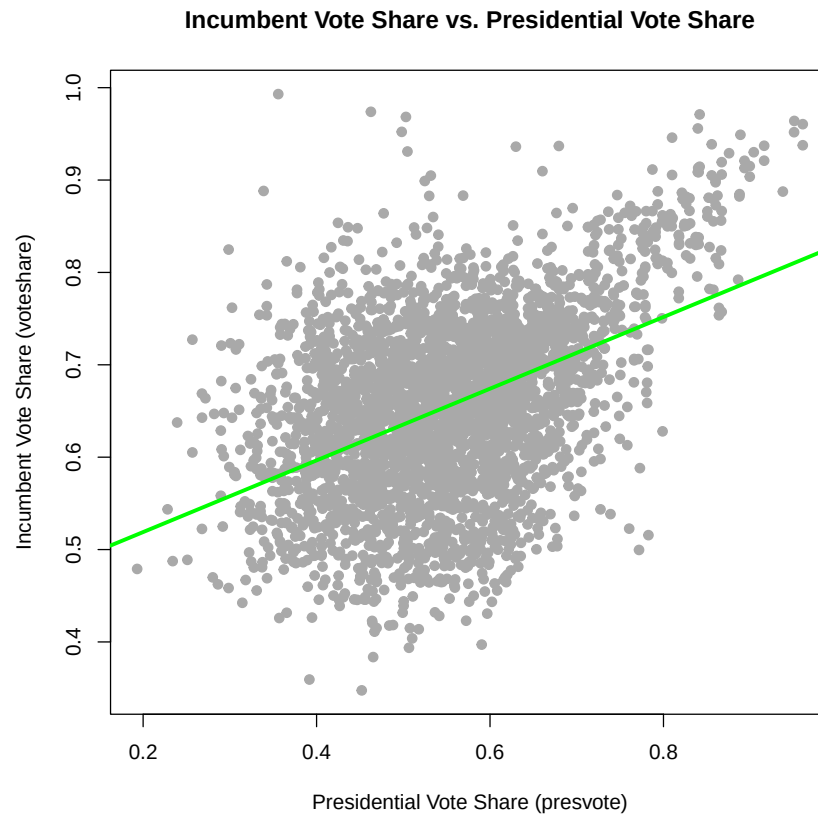


Figure 3: Scatterplot of incumbent vote share vs. presidential vote share of the incumbent's party

Computed in R as:

```
1 #Q3.2: Scatterplot with regression line
2 pdf("scatterplot_model3.pdf")
3 plot(inc_sub$presvote, inc_sub$voteshare,
4       main = "Incumbent Vote Share vs. Presidential Vote Share",
5       xlab = "Presidential Vote Share (presvote)",
6       ylab = "Incumbent Vote Share (voteshare)",
7       pch = 19, col = "darkgray")
8 abline(model3, col = "green", lwd = 3)
9 dev.off()
```

3. Write the prediction equation.

**Answer:** The prediction equation indicates that for each one percentage point increase in the presidential vote share of the incumbent's party, the incumbent's own vote share is expected to increase by approximately 0.388 percentage points. The prediction equation reads as follows.

$$\widehat{\text{voteshare}} = 0.441 + 0.388 \cdot \text{presvote}$$

## Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

**Answer:** To assess the relationship between unexplained variation in incumbent vote share and presidential vote share after accounting for spending differences, a regression was estimated using the residuals from Questions 1 and 2.

Table 4: Regression of Residuals: Incumbent Vote Share on Presidential Vote Share

|                         | <i>Dependent variable:</i>  |
|-------------------------|-----------------------------|
|                         | residuals_model1            |
| residuals_model2        | 0.257***<br>(0.012)         |
| Constant                | -0.000<br>(0.001)           |
| Observations            | 3,193                       |
| R <sup>2</sup>          | 0.130                       |
| Adjusted R <sup>2</sup> | 0.130                       |
| Residual Std. Error     | 0.073 (df = 3191)           |
| F Statistic             | 476.975*** (df = 1; 3191)   |
| <i>Note:</i>            | *p<0.1; **p<0.05; ***p<0.01 |

Computed in R as:

```

1 #Q4.1: Run regression of residuals
2 model4 <- lm(residuals_model1 ~ residuals_model2)
3 summary(model4)

```

2. Make a scatterplot of the two residuals and add the regression line.

**Answer:** The scatterplot below visualises the relationship between the residuals from the first two models, highlighting how unexplained variation in presidential vote share corresponds to unexplained variation in incumbent vote share after controlling for spending differences.

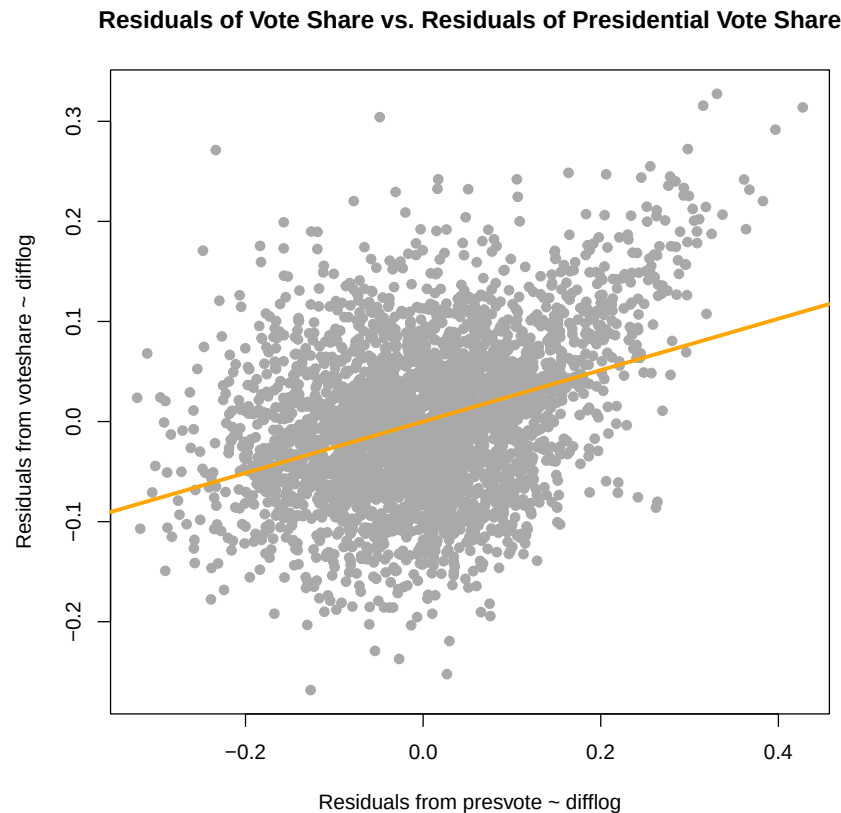


Figure 4: Scatterplot of residuals: incumbent vote share vs. presidential vote share, controlling for spending differences

Computed in R as:

```
1 #Q4.2: Scatterplot with regression line
2 pdf("scatterplot_model4.pdf")
3 plot(residuals_model2, residuals_model1,
4       main = "Residuals of Vote Share vs. Residuals of Presidential Vote
5       Share",
6       xlab = "Residuals from presvote ~ difflog",
7       ylab = "Residuals from voteshare ~ difflog",
8       pch = 19, col = "darkgray")
9 abline(model4, col = "orange", lwd = 3)
dev.off()
```

3. Write the prediction equation.

**Answer:** The prediction equation indicates that, after controlling for spending differences, a one-unit increase in the residual presidential vote share is associated with a 0.257 unit increase in the residual incumbent vote share. The intercept is effectively zero, suggesting no systematic bias when both predictors are centered around their model expectations. The prediction equation reads as follows.

$$\widehat{\text{resid}_{\text{voteshare}}} = -0.000 + 0.257 \cdot \text{resid}_{\text{presvote}}$$

## Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

**Answer:** A multiple linear regression was estimated to assess how both `difflog` and `presvote` jointly affect the incumbent's `voteshare`.

Table 5: Regression of Incumbent Vote Share on Spending Difference and Presidential Vote Share

|                         | <i>Dependent variable:</i>  |
|-------------------------|-----------------------------|
|                         | voteshare                   |
| difflog                 | 0.036***<br>(0.001)         |
| presvote                | 0.257***<br>(0.012)         |
| Constant                | 0.449***<br>(0.006)         |
| Observations            | 3,193                       |
| R <sup>2</sup>          | 0.450                       |
| Adjusted R <sup>2</sup> | 0.449                       |
| Residual Std. Error     | 0.073 (df = 3190)           |
| F Statistic             | 1,302.947*** (df = 2; 3190) |
| <i>Note:</i>            | *p<0.1; **p<0.05; ***p<0.01 |

Computed in R as:

```

1 #Q5.1: Run multiple linear regression
2 model5 <- lm(voteshare ~ difflog + presvote, data = inc_sub)
3 summary(model5)

```

2. Write the prediction equation.

**Answer:** The prediction equation below expresses the expected incumbent vote share as a function of both predictors.

$$\widehat{\text{voteshare}} = 0.449 + 0.036 \cdot \text{difflog} + 0.257 \cdot \text{presvote}$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

**Answer:** The coefficient on `presvote` in this model is exactly the same as the slope from the regression in Question 4. This happens because both models are asking the same underlying question: how does presidential vote share affect incumbent vote share, once we've already accounted for the influence of spending differences?

In Question 4, we removed the effect of spending by working with residuals — that is, we looked at what's left over in both `voteshare` and `presvote` after subtracting out the part explained by `difflog`. Then we ran a regression on those leftover pieces.

In Question 5, we included both `difflog` and `presvote` directly in the same model. This gives us the effect of `presvote` while holding `difflog` constant — which is mathematically the same as regressing the residuals from Question 4.

So even though the two approaches look different, they're measuring the same relationship in slightly different ways. That's why the coefficient on `presvote` is identical in both cases.